

Bi-directional Dynamic Programming for the k -color Shortest Path Problem

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Abstract

The k -color Shortest Path Problem (k -CSPP) is an NP-hard optimization challenge that involves finding the minimum-cost path between a source and a destination in an edge-colored graph while using at most k distinct colors. This report proposes and evaluates a bounded bi-directional dynamic programming (BDP) approach to solve the k -CSPP as an alternative to existing mono-directional dynamic programming methods. Computational experiments demonstrate that BDP maintains competitive execution times on standard random and grid graphs. On newly generated labyrinth instances specifically designed to challenge existing algorithms, BDP significantly outperforms both mono-directional DP and Valid Inequalities Branch-and-Cut methods by significantly reducing the required search space. The report confirms that layered-based digraphs remain challenging for dynamic programming techniques due to severe combinatorial explosion.

1 Introduction

The project is focused on the optimization of the k -color Shortest Path Problem [1] (k -CSPP), which is a variant of the classic Shortest Path Problem [2] (SPP) that we know can be solved in polynomial time.

This problem arises in many real-world applications: the design of reliable telecommunication networks, multimodal transportation, obstacle minimization in robotics, and wavelength-routed optical networks.

The k -CSPP consists of finding a shortest path in a weighted edge-colored graph, where the maximum number of distinct colors that can appear in the path is fixed to be k . Considering a digraph $D = (\mathcal{V}, A)$ and a set of colors C , every arc (i, j)

has a positive cost d_{ij} and color $c_{ij} \in C$. Given a source node s and a target node t ($s, t \in \mathcal{V}$), alongside a positive integer k , the k-CSPP consists of finding an (s, t) -path of minimum cost while using at most k distinct arc colors.

The first formal definition of the k-color Shortest Path Problem was presented by Ferone et al. (2019) [1].

Existing methodologies include a Branch & Bound algorithm (B&B) [1], a dynamic programming approach with an A*-like technique (DP) [3], a near-optimal heuristic called color-constrained Dijkstra (CCDA) with an instance reduction graph procedure (GRA) [4], and a Branch & Cut procedure with new Valid Inequalities [5] that seems to perform better on new layered digraph generated challenging instances.

The goal of this project is to evaluate a possible alternative algorithm to face the k-CSPP which is an NP-Hard combinatorial problem by extending the Dynamic Programming and using a Bounded Bi-directional Dynamic Programming (BDP) approach (Righini et al. 2006 [6]).

We will evaluate the limits of this technique across different benchmark instances already used in the literature (Table 1) and on a newly generated set of instances designed to challenge existing solutions (Table 2).

The proposed BDP approach maintained competitive execution times on standard random and grid digraphs (Table 3), while on the newly generated labyrinth instances, BDP outperformed both mono-directional DP and Valid Inequalities Branch-and-Cut methods (Table 5).

The evaluation successfully mapped the boundaries of the approach, confirming that layered-based digraphs remain challenging for dynamic programming techniques due to the combinatorial explosion (Table 4).

2 Problem Formulation

The mathematical model, based on a flow formulation, is presented here to clarify the characteristics of the problem ([1]).

Let G be a digraph $D = (\mathcal{V}, A)$, with $|\mathcal{V}|$ nodes and $|A|$ arcs. Given two functions $C : A \rightarrow \mathbb{Z}^+$ and $d : A \rightarrow \mathbb{R}^+$: the first is an arc-coloring function, where the color of the arc a is defined as a positive integer $C(a)$, $\forall a \in A$. The second is a non-negative distance function defined for each arc $a \in A$. We define $\mathcal{C} = \cup_{a \in A} \{C(a)\}$ as the set containing all the colors of G . For each color $h \in \mathcal{C}$, we define $A_h = \{a \in A \text{ s.t. } C(a) = h\}$ as the set containing all the arcs with colors equal to h .

The k-CSPP consists of finding a minimum distance path from a source node s to a destination node t , with $s, t \in \mathcal{V}$, such that the number of different colors traversed on the path does not exceed k .

First, let $x_{uv} \forall (u, v) \in A$, be a binary decision variable to represent whether arc (u, v) belongs to the solution ($x_{uv} = 1$) or not ($x_{uv} = 0$). Second, $\forall h \in \mathcal{C}$, let y_h be a binary decision variable to determine whether a color h is in the solution ($y_h = 1$) or not ($y_h = 0$).

$$(\text{FFP}) \min \sum_{(u,v) \in A} d_{uv} x_{uv} \tag{1}$$

$$\text{s.t.} \quad \sum_{(v,u) \in A} x_{vu} - \sum_{(u,v) \in A} x_{uv} = \begin{cases} -1 & \text{if } u = s \\ +1 & \text{if } u = t \\ 0 & \text{otherwise,} \end{cases} \quad \forall u \in \mathcal{V}, \quad (2)$$

$$x_{uv} \leq y_{c(u,v)}, \quad \forall (u,v) \in A, \quad (3)$$

$$\sum_{h \in \mathcal{C}} y_h \leq k, \quad (4)$$

$$x_{uv} \in \{0, 1\}, \quad \forall (u,v) \in A, \quad (5)$$

$$y_h \in \{0, 1\}, \quad \forall h \in \mathcal{C}. \quad (6)$$

The objective function (1) minimizes the cost of the (s, t) -path. The constraints in (2) are the flow conservation constraints. The constraints in (3) enable the passage of the flow only on arcs whose color is selected. Constraint (4) is used to limit the maximum number of usable colors to k . Finally, constraints (5) and (6) are used to define the Boolean domain of the decision variables.

3 Solution Approach

In this section, we propose a bi-directional dynamic programming approach to solve the k-CSPP, as an alternative to the mono-directional DP [3] already used to solve this problem. We will start by discussing the mono-directional DP approach and its complexity, and then we will detail the bi-directional solution.

3.1 Dynamic Programming Solution

Let P_{si} be a path in D connecting s with a generic node i , and let $L_i = (d_i, C_i, P_{si})$ be a label associated with P_{si} , where d_i is the total distance of the path and C_i is the set of distinct colors of the path P_{si} . Let $D(i)$ be defined as the set of all the labels L_i associated with the different paths connecting s to i .

The exploration is based on the label correcting¹ technique; the DP explores the solution space to extend the paths under construction by analyzing the set of their labels. The extension operation of a given path P_{si} to a path P_{sh} is the result of the concatenation $(P_{si}, [i, h])$ with $[i, h] \in A \setminus P_{si}$. The source node has an initial label $L_s = (0, \emptyset, [s])$, and starting from a generic label $L_i = (d_i, C_i, P_{si})$, $\forall j \in \mathcal{V} : [i, j] \in A$ new labels L_j are generated. The distance of the path P_{sj} is defined as $d_j = d_i + w_{ij}$ and the set of colors traversed is $C_j = C_i \cup \{C([i, j])\}$.

The computational effort of a DP algorithm is related to the number of explored labels. With no specific extraction policy, in the worst case, it is equal to the number of feasible k-colored paths from s (source) to t (destination).

Considering A* as the extraction policy, where the extracted label for a generic node i is the one that presents the smallest value $(d_i + ssp_i)$, with ssp_i being the simple shortest path (without color constraints) from the current node i to the target node t . We can then consider an upper bound of the number of labels generated for

¹label correcting maintains mutable labels that are progressively updated and pruned via dominance rules and bounding constraints until the search space converges

each node as:

$$N(k) = \sum_{l=1}^k \binom{|\mathcal{C}|}{l}$$

Consequently, an upper bound for the complexity of DP with the A* strategy is $O((|V| - 2) \cdot N(k)^2)$ [3].

3.2 Elementary and Dominance

The (k -CSPP) is defined on a digraph where the distance function assigns a non-negative distance to each arc, therefore, the shortest path will naturally be a simple (elementary) path.

In order to prune the search space efficiently, a dominance rule is applied to the labels generated during the dynamic programming execution.

Let $L_i = (d_i, C_i)$ and $\hat{L}_i = (\hat{d}_i, \hat{C}_i)$ be two distinct state labels representing paths from the source node s to a given node i , where d_i represents the total path distance and C_i represents the set of traversed colors.

According to Ferone et al. (2021) [3], the label L_i dominates $\hat{L}_i$ if the following conditions hold:

1. $d_i \leq \hat{d}_i$
2. $C_i \subseteq \hat{C}_i$

with at least one condition being strictly satisfied.

Furthermore, it is guaranteed [3] that extending a dominated path $\hat{P}_{si}$ will only ever yield a newly dominated path. Because the problem assumes non-negative arc weights, any potential cycles formed during the extension of the dominant path P_{si} can be naturally truncated.

3.3 Bi-Directional Solution

In the bi-directional dynamic programming (BDP) approach, the labels are extended both forward from node s to its successors and backward from node t to its predecessors. Backward states, extension functions, and dominance rules are symmetrical. Once the extension is completed, the forward and backward paths are joined to produce complete paths from node s to node t , ensuring that the final path contains no cycles nor violates resource constraints. (Righini et al. (2006) [6])

In Algorithm 1, we use bounding to recognize and fathom the states that cannot produce optimal solutions, stopping the extension of forward and backward paths in order to reduce the number of generated states while preserving the guarantee that the optimal solution will be found.

This limitation, ensures that the bi-directional algorithm doesn't produce as many labels as the mono-directional one.

Algorithm 1 Bounded Bi-directional Dynamic Programming

```
1: // Initialization //
2:  $(A_r, UB) \leftarrow \text{CCDA}(\mathcal{V}, A, C, s, t, k_{max})$ 
3:  $\Gamma_s^{fw} \leftarrow \{L_s(0, \emptyset, s)\}$ 
4:  $\Gamma_t^{bw} \leftarrow \{L_t(0, \emptyset, t)\}$ 
5:  $\Gamma_i^{fw} \leftarrow \emptyset, \forall i \in \mathcal{V}, i \neq s$ 
6:  $\Gamma_i^{bw} \leftarrow \emptyset, \forall i \in \mathcal{V}, i \neq t$ 
7:  $E \leftarrow \{s, t\}$ 
8: // Search //
9: while  $E \neq \emptyset$  do
10: // Node selection //
11: Select  $i \in E$ ;  $E \leftarrow E \setminus \{i\}$ 
12: // Forward extension //
13: Sort  $\bar{\Gamma}_i^{fw}$  by non-decreasing number of colors
14: for all  $L_i = (i, d_i, C_i) \in \bar{\Gamma}_i^{fw}$  do
15:   for all  $j \in \mathcal{V} : (i, j) \in A_r$  do
16:      $L_j \leftarrow \text{Extend}(d_i + w_{ij}, C_i \cup \{C([i, j])\}, k_{max}, h_j^{fw}, UB)$ 
17:      $\Gamma_j^{fw} \leftarrow \text{AddLabel}(\Gamma_j^{fw}, L_j)$ 
18:     if  $\bar{\Gamma}_j^{fw} \neq \emptyset$  then
19:        $E \leftarrow E \cup \{j\}$ 
20:     end if
21:   end for
22: end for
23: // Backward extension //
24: Sort  $\bar{\Gamma}_i^{bw}$  by non-decreasing number of colors
25: for all  $L_i = (i, d_i, C_i) \in \bar{\Gamma}_i^{bw}$  do
26:   for all  $j \in \mathcal{V} : (j, i) \in A_r$  do
27:      $L_j \leftarrow \text{Extend}(d_i + w_{ji}, C_i \cup \{C([j, i])\}, k_{max}, h_j^{bw}, UB)$ 
28:      $\Gamma_j^{bw} \leftarrow \text{AddLabel}(\Gamma_j^{bw}, L_j)$ 
29:     if  $\bar{\Gamma}_j^{bw} \neq \emptyset$  then
30:        $E \leftarrow E \cup \{j\}$ 
31:     end if
32:   end for
33: end for
34: end while
35:  $\text{Join}(\mathcal{V}, \Gamma^{bw}, \Gamma^{fw})$ 
```

The BDP Algorithm 1 is a variation of the method proposed by Righini et al. (2006) [6]. First, a graph reduction A_r (GRA [4]) and an upper bound are calculated (Line 2). For each node $i \in \mathcal{V}$, we indicate with Γ_i the set of labels associated with the node, with $\bar{\Gamma}_i \subseteq \Gamma_i$ being the subset of labels not extended so far. E is the set of nodes to be explored; *Extend* is the extension procedure from i to j . This procedure checks

the color constraints and applies bounding for fathoming. *AddLabel* is the procedure that inserts the new label L into the set Γ by applying dominance rules (Algorithm 2).

Algorithm 2 Dominance check & Adding Label (*AddLabel*)

Require: Γ_j, L_j

- 1: **if** L_j **is not dominated by any** label $L'_j \in \Gamma_j$ **then**
 - 2: Remove from Γ_j all labels L'_j that are dominated by L_j .
 - 3: $\Gamma_j \leftarrow \Gamma_j \cup \{L_j\}$
 - 4: **end if**
-

3.4 Bounding for fathoming

In our experiment, we calculated an Upper Bound (UB) using the CCDA [4] and used it to fathom infeasible labels.

As Lower Bound (LB) we check if the cost of the label plus the lower bound calculated as the shortest path to the target node (t or s) without constraints is greater than or equal to the calculated upper bound (UB), that is, $Cost(Label) + LB \geq UB$.

The Extend (3) function checks :

- Whether the new distance $d_j + h_j \geq UB$ (h_j is the pre-calculated shortest path from node j to the node t , or s for backward, without color constraints) . (Lines 2-3)
- The feasibility of the new extended label by checking if the number of colors is less than or equal to k_{max} . (Lines 4-5)

Algorithm 3 Extension of the label (*Extend*)

Require: $d_j, C_j, k_{max}, h_j, UB$

- 1: $H \leftarrow UB/2.0$
 - 2: **if** $d_j + h_j \geq UB$ **or** $d_j > H$ **then**
 - 3: **return** $L_j \leftarrow \emptyset$
 - 4: **else if** $|C_j| \leq k_{max}$ **then**
 - 5: **return** $L_j \leftarrow (d_j, C_j)$
 - 6: **else**
 - 7: **return** $L_j \leftarrow \emptyset$
 - 8: **end if**
-

3.5 Half-Way Point

To avoid repetition of the calculated labels, in bi-directional programming it is important to find a half-way point to halt exploration, this let to reduce the numbers of explorations, but it is important even to guarantee that the remaining part of the path is generated in the other direction so that no optimal solution is lost.

Usually it is suggested to use the most binding critical resource as the critical resource to define the halfway meeting point and it must be a strictly additive scalar (like time, capacity or distance). In the k-CSPP the 'set of used colors' are non-additive. This prevents us from defining a safe $k/2$ stopping criterion without losing optimal solutions.

Therefore we define our half-way point denoted H as half of the cost of the Upper Bound calculated with the CCDA algorithm [4]. It is implemented in the Extend function 3 where we check if the d_j current distance cost of the label achieved half of the cost path. (Line 2-3).

A drawback of this choice is that if the CCDA cannot find any upper bound, the bi-directional approach will double its exploration w.r.t the mono-DP approach. A future improvement could try to mitigate this issue using alternative heuristics or bounding techniques such as arc-bounding.

3.6 Join

The final phase of the bi-directional dynamic programming algorithm connects forward and backward paths at an arc (i, j) (Algorithm 4).

For each valid arc $(i, j) \in A_r$, the forward label list at the source node (Γ_i^{fw}) and the backward label list at the destination node (Γ_j^{bw}) are sorted by distance in ascending order (Line 3).

First, it checks if the sum of the best forward distance, the arc cost w_{ij} , and the best backward distance exceeds the known incumbent; if so, the entire arc is safely discarded (Lines 5-7).

During the nested evaluation of individual label pairs, if the combined distance ($d^{fw} + w_{ij} + d^{bw}$) exceeds the incumbent, the inner loop immediately halts (Lines 8-10), further pruning the search space.

Finally, the algorithm verifies the color feasibility of the merged path (Line 12). This is performed by checking whether the cardinality of the union of the forward color set, the backward color set, and the specific color of the connecting arc ($|C^{fw} \cup C^{bw} \cup \{c_{ij}\}|$) is less than or equal to k_{max} .

If the merged path is strictly feasible and improves the current incumbent, the value is updated (Line 13).

4 Computational results

In this section, we present numerical experiments performed on a MacBook Pro M5, 24 GB DDR3 RAM with MacOS Tahoe. We use Julia 1.12.6 with JuMP package to implement models for CPLEX 22.1 configured with one thread. The time limit for each instance is set to 600 s.

4.1 Test instances

We used the benchmark instances (grid and random digraphs) from the literature (Ferone et al., 2019) (Table 1), a set of hard instances of layered-based digraphs (Castro et al., 2024) (Table 2) where k is set as the minimum number of colors allowing an

Algorithm 4 Join

```
1: for all  $(i, j) \in A_r$  do
2:   Let  $w_{ij}$  be the cost and  $c_{ij}$  the color of arc  $(i, j)$ 
3:   Sort  $\Gamma_i^{fw}$  and  $\Gamma_j^{bw}$  by non-decreasing distance
4:   for all  $L^{fw} = (i, d^{fw}, C^{fw}) \in \Gamma_i^{fw}$  do
5:     if  $d^{fw} + w_{ij} + \min(d^{bw}) \geq incumbent$  then
6:       break
7:     end if
8:     for all  $L^{bw} = (j, d^{bw}, C^{bw}) \in \Gamma_j^{bw}$  do
9:       if  $d^{fw} + w_{ij} + d^{bw} \geq incumbent$  then
10:        break
11:       end if
12:       if  $|C^{fw} \cup C^{bw} \cup \{c_{ij}\}| \leq k_{max}$  then
13:          $incumbent \leftarrow d^{fw} + w_{ij} + d^{bw}$ 
14:       end if
15:     end for
16:   end for
17: end for
18: return  $incumbent$ 
```

(s, t) -path, obtained after solving the MCPP (Minimum Color Path Problem). We also create a new set of 5 instances (Table 2). These instances have two disjoint paths from s (source) to destination t , where the first path is a trivial path but with high cost (Figure 1). The second path is a wide, deep grid of near-zero-cost arcs where the colors of the arcs at each level of the grid introduces one new color and retires one old one, so exactly k_{max} colors are available at any level but $2 \cdot k_{max}$ distinct colors appear across the full graph. Any complete path through it therefore requires more than k_{max} colors. The instance is calibrated so that the second path looks attractive on cost alone, forcing each algorithm to discover its infeasibility through exploration.

To enhance efficiency, all the instances are preprocessed using the graph reduction algorithm, which eliminates arcs proven to be unnecessary for the optimal solution based on feasible solutions obtained through the CCDA heuristic (Cerrone & Russo, 2023) [4].

4.2 Results for DP, BDP, CPLEX, Valid Inequalities

In Table 3, we show the numerical results for the benchmark instances for random and grid digraph using the Dynamic Programming procedure with the A^* extraction policy (Ferone et al. 2019 [3]), CPLEX solver with the CCDA [4] heuristic solution as a cutoff value, our implementation of the FFP model restricted to valid inequalities (17), (18), and (19) as proposed by Castro et al. (2024)[5]) and finally our BDP solution.

The table also reports the CCDA upper bound (ub) calculated for every *instance*. The computational time (cpu) is reported in seconds. The (-) indicates the algorithm reached the time limit or ran out of memory. Each group (R1, R3, G1, G4) is separated

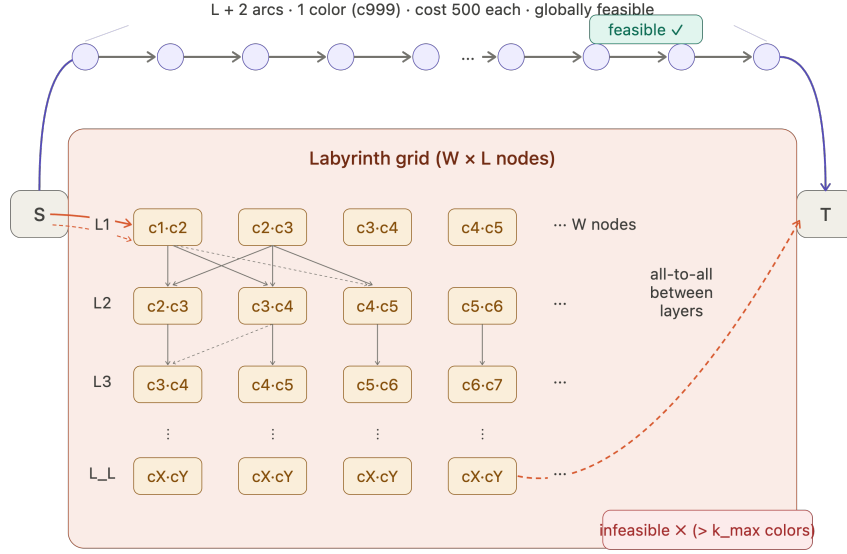


Fig. 1: Example of a labyrinth instance: two disjoint paths from s to t , where the grid path introduces one new color per layer.

by horizontal line, to calculate the statistics values Average, Median, Max and Min of every group.

The same legend has been used for layered-based digraph (Table 4) and the new-generated instances (Table 5).

As already known from the literature [3] the DP algorithm performs well on average in the random/grid graph, there is some exception on some grid instance G1-27004 and G4-27000.

The BDP approach achieves similar performance on the random digraph, showing a better median values for the grids groups (G1 and G4). The layered-based digraphs (Table 4) both the DP and BDP algorithms suffer from combinatorial explosion due to the topology of this type of the graph and the k which is set as the MCPP, this forces the process to explore all combinations to the ends, pruning very few labels at each time.

With the newly generated (Table 5) instances, because the labyrinth (Figure 1) introduces only 1 new color per layer, the DP must travel $k_{max} + 1$ layers deep into the grid before a path finally accumulates enough colors to trigger the cutoff. BDP enters the labyrinth from both sides simultaneously and only does half of the work on the optimal path.

The IBM CPLEX with the fixed ub , turns out to be a bit slower than DP and BDP in the random graph on average, but has results in a fastest execution on grid digraph. Regarding the layered-based digraph were all solved to optimality with an average

Table 1: Details about the benchmark instances for random and grid digraphs (Ferone et al., 2019).

Random						Grid					
<i>inst</i>	$ V $	$ A $	$ R(A) $	k	opt	<i>inst</i>	$ V $	$ A $	$ R(A) $	k	opt
R1-27190	75 000	750 000	16	8	242	G1-27001	10 000	39 600	210	197	6 233
R1-27191	75 000	750 000	39	6	201	G1-27003	10 000	39 600	444	196	6 203
R1-27195	75 000	750 000	13	6	152	G1-27004	10 000	39 600	272	195	6 375
R1-27197	75 000	750 000	12	6	253	G1-27007	10 000	39 600	223	198	6 197
R1-27199	75 000	750 000	48 383	5	333	G1-27008	10 000	39 600	215	191	6 193
R1-27200	75 000	750 000	143	6	236	G1-27009	10 000	39 600	255	196	6 183
R3-27004	75 000	750 000	29	6	246	G4-27000	62 500	498 500	859	767	23822
R3-27005	75 000	750 000	25	6	198	G4-27002	62 500	498 500	1209	770	23746
R3-27007	75 000	750 000	38	6	231	G4-27003	62 500	498 500	910	767	23610
R3-27008	75 000	750 000	15	7	196	G4-27004	62 500	498 500	1159	774	23904
R3-27010	75 000	750 000	178	5	246	G4-27005	62 500	498 500	883	773	23995
R3-27012	75 000	750 000	28	7	245	G4-27006	62 500	498 500	1519	762	23940

Table 2: Details about the benchmark instances for layered-based digraphs (Castro et al., 2024) and the newly generated instances.

Layered-based (Castro et al., 2024)						Newly generated instances					
<i>inst</i>	$ V $	$ A $	$ R(A) $	k	opt	<i>inst</i>	$ V $	$ A $	$ R(A) $	k	opt
GP1-3	152	1 420	1 420	7	5 953	N-6	528	9 667	9 640	4	260.0
GP1-4	152	1 420	1 420	7	5 056	N-8	1 088	30 697	30 660	20	360.0
GP2-22	152	1 447	1 447	7	6 211	N-9	1 438	54 517	54 480	30	360.0
GP2-23	152	1 439	1 430	7	5 897	N-10	2 298	110 147	110 100	40	460.0
GP2-40	152	1 440	1 434	7	6 845	N-11	3 358	194 577	194 520	50	560.0

time of 26.46 seconds. The new instances have been resolved, but as the graph grows in the depth, CPLEX starts to perform badly, because there are exactly k_{max} colors per layer, the fractional flow passing through any specific color h is exactly $1/k_{max}$. This forces the LP relaxation to split the unit flow of 1.0 across the width of the grid.

Concerning the B&C strategies with Valid Inequalities, as is known from the literature [5] it performs well on average both random, grid digraph and layered digraph. The strongest valid inequalities state that the sum of the flow on arcs of color h in a certain cut or non-reachable subset must be $\leq y_h$ (the color activation variable). CPLEX perfectly satisfies all these complex cuts by setting the fractional variable $y_h = 1/k_{max}$. Once CPLEX realizes the variables are fractional (0.2) and not binary (0 or 1), it initiates the Branch-and-Bound phase. However, because the grid contains millions of perfectly symmetrical paths, there is no "best direction" to branch on.

The solver gets lost exploring an exponentially large search tree of identical fractional values until the timeout.

5 Conclusions and future works

This project proposed and evaluated a bounded bi-directional dynamic programming (BDP) [6] approach to solve the NP-hard k -color Shortest Path Problem (k -CSPP). By extending labels simultaneously from the source and destination nodes and utilizing a half-way stopping criterion based on an upper bound heuristic, the BDP algorithm effectively limits redundant label generation.

Computational experiments (Table 3) demonstrate that BDP maintains competitive execution times on standard random and grid digraphs. Furthermore, on newly generated labyrinth instances BDP significantly outperformed (Table 5) both mono-DP and Valid Inequalities by reducing the required search space and quickly identifying path infeasibilities. Despite these improvements, layered-based digraphs remain exceptionally challenging (Table 4) for dynamic programming techniques due to severe combinatorial explosion.

Future improvements could include integrating other bounding techniques like State Space Relaxation [7] or Decremental State Space Relaxation [6] by forcing the algorithm to only track a subset of the most frequently used colors. This could help to reduce the exploration, even though the non-additive property of the colors is an important limit to take into account.

Additionally, the development of a hybrid approach that combines bi-directional dynamic programming with Valid Inequalities. Since Valid Inequalities [5] demonstrated strong performance and stability across layered digraphs (Table 4), integrating them to compute tighter bounds could mitigate the combinatorial explosion that limits DP-based techniques on those specific topologies.

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Table 3: Numerical results for DP, BDP, CPLEX, and Valid Inequalities for random and grid digraphs (Ferone et al., 2019)

<i>Instance</i>	CCDA	DP	BDP	CPLEX	Inequalities
	<i>ub</i>	<i>cpu</i>	<i>cpu</i>	<i>cpu</i>	<i>cpu</i>
R1-27190	242	0.0012	0.0125	0.5218	0.0196
R1-27191	201	0.0011	0.0136	0.1089	0.0175
R1-27195	152	0.0010	0.0057	0.1092	0.0161
R1-27197	253	0.0011	0.0061	0.0952	0.0146
R1-27199	333	0.1189	0.1373	2.4469	2.9082
R1-27200	236	0.0014	0.0062	0.1265	0.0184
Average	236.2	0.0208	0.0302	0.5681	0.4991
Median	239.0	0.0011	0.0094	0.1179	0.0180
Max	333.0	0.1189	0.1373	2.4469	2.9082
Min	152.0	0.0010	0.0057	0.0952	0.0146
R3-27004	246	0.0011	0.0060	0.1187	0.0173
R3-27005	198	0.0011	0.0058	0.1169	0.0169
R3-27007	231	0.0011	0.0057	0.1069	0.0175
R3-27008	196	0.0013	0.0058	0.1366	0.0166
R3-27010	246	0.0012	0.0061	0.1186	0.0180
R3-27012	245	0.0011	0.0058	0.1226	0.0177
Average	227.0	0.0011	0.0059	0.1201	0.0173
Median	238.0	0.0011	0.0058	0.1187	0.0174
Max	246.0	0.0013	0.0061	0.1366	0.0180
Min	196.0	0.0011	0.0057	0.1069	0.0166
G1-27001	6233	0.0002	0.0007	0.0073	0.0018
G1-27003	6203	0.0026	0.0011	0.0135	0.0045
G1-27004	6375	45.2761	0.0016	0.0082	0.0027
G1-27007	6197	0.0007	0.0008	0.0072	0.0017
G1-27008	6193	0.0003	0.0008	0.0074	0.0017
G1-27009	6183	0.0006	0.0009	0.0080	0.0024
Average	6230.7	7.5468	0.0010	0.0086	0.0025
Median	6200.0	0.0007	0.0009	0.0077	0.0021
Max	6375.0	45.2761	0.0016	0.0135	0.0045
Min	6183.0	0.0002	0.0007	0.0072	0.0017
G4-27000	23822	16.3234	0.0267	0.1595	0.0210
G4-27004	23914	0.1367	127.8307	0.1238	0.0376
G4-27005	23995	0.0273	0.0499	0.1397	0.0279
G4-27006	23940	-	11.3697	0.1356	0.0326
G4-27007	23953	0.0928	0.0180	0.1225	0.0215
G4-27009	23798	0.3283	0.0162	0.1049	0.0185
Average	23903.7	3.3817	23.2185	0.1310	0.0265
Median	23927.0	0.1367	0.0383	0.1297	0.0247
Max	23995.0	16.3234	127.8307	0.1595	0.0376
Min	23798.0	0.0273	0.0162	0.1049	0.0185

Table 4: Numerical results for DP, BDP, CPLEX, and Valid Inequalities for layered-based digraphs (Castro et al., 2024)

<i>Instance</i>	<u>CCDA</u>	<u>DP</u>	<u>BDP</u>	<u>CPLEX</u>	<u>Valid Inequalities</u>
	<i>ub</i>	<i>cpu</i>	<i>cpu</i>	<i>cpu</i>	<i>cpu</i>
GP1-3	-	-	-	26.9545	2.3358
GP1-4	-	-	-	2.9353	1.6920
GP2-22	51145	-	-	23.8463	1.9515
GP2-23	27388	-	-	19.4913	2.3121
GP2-40	27264	-	-	59.0637	13.7712
Average	35265.7	-	-	26.4582	4.4125
Median	27388.0	-	-	23.8463	2.3121
Max	51145.0	-	-	59.0637	13.7712
Min	27264.0	-	-	2.9353	1.6920

Table 5: Numerical results for DP, BDP, CPLEX, and Valid Inequalities (New Instances)

<i>Instance</i>	<u>CCDA</u>	<u>DP</u>	<u>BDP</u>	<u>CPLEX</u>	<u>Valid Inequalities</u>
	<i>ub</i>	<i>cpu</i>	<i>cpu</i>	<i>cpu</i>	<i>cpu</i>
N-6	260	0.0261	0.0001	0.9947	1.6116
N-8	360	-	0.0001	6.1689	1.9709
N-9	360	-	0.0001	5.5231	5.6575
N-10	460	-	0.0002	40.1596	16.6383
N-11	560	-	0.0003	377.3205	42.2813
Average	400.0	0.0261	0.0002	86.0334	13.6319
Median	360.0	0.0261	0.0001	6.1689	5.6575
Max	560.0	0.0261	0.0003	377.3205	42.2813
Min	260.0	0.0261	0.0001	0.9947	1.6116

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